

# Six-Point Trefoil Probability

Assume the same model as before: six i.i.d. uniform points on  $S^2$ , connected in a fixed cyclic order by straight chords.

$$\Pr(\text{trefoil}) = 1/(16\pi^2)$$

Numerically,

$$1/(16\pi^2) = 0.006332573977646\dots$$

so the probability is approximately

$$0.6332574\%$$

## The derivation is as follows.

For six points on a sphere, the convex hull is almost surely one of two combinatorial types:

| Convex-hull type        | Trefoil cyclic orders among the 60 unoriented cycles |
|-------------------------|------------------------------------------------------|
| Octahedral              | 1                                                    |
| Biaugmented tetrahedral | 0                                                    |

The finite cycle classification follows from Calvo's six-stick knot criterion, which distinguishes trefoils using signed triangle piercings: [Calvo's classification](#).

Thus

$$\Pr(\text{trefoil}) = (1/60) \Pr(\text{octahedral hull}).$$

Fix one vertex as the north pole and stereographically project the other five points to the plane. Their density is

$$f(x,y) = 1/[\pi(1+x^2+y^2)^2].$$

Let  $X$  be the probability mass, under this density, of the triangle formed by three projected points. A fixed vertex has degree 3 precisely when one of the  $C(5,3) = 10$  possible triangles contains the other two, so

$$x_3 = 10 E[X^2].$$

The irregular hull has exactly two degree-3 vertices, while the octahedral hull has none. Therefore

$$\Pr(\text{octahedral hull}) = 1 - 30 E[X^2].$$

To evaluate  $E[X^2]$ , use the one-form

$$\omega = (x dy - y dx)/(1+x^2+y^2), \quad d\omega = 2 dx dy/(1+x^2+y^2)^2.$$

For a triangle with vertices  $a, b, c$ ,

$$X^2 = [I(a,b) + I(b,c) + I(c,a)]^2/(4\pi^2),$$

where

$$I(a,b) = \int_a^b \omega.$$

Reflection symmetry makes the cross terms vanish. Writing the line through  $a, b$  as

$$a = p n + u t, \quad b = p n + v t, \quad v > u,$$

and then substituting  $u = \sqrt{1+p^2} \tan \alpha$ ,  $v = \sqrt{1+p^2} \tan \beta$ , gives

$$E[X^2] = (3/(4\pi^2)) (2/\pi) (4/15) (\pi^3/12 - 5\pi/16).$$

Hence

$$E[X^2] = 1/30 - 1/(8\pi^2).$$

Therefore

$$\text{Pr}(\text{octahedral hull}) = 1 - 30(1/30 - 1/(8\pi^2)) = 15/(4\pi^2),$$

and finally

$$\text{Pr}(\text{trefoil}) = (1/60)(15/(4\pi^2)) = 1/(16\pi^2).$$

If you meant great-circle arcs lying on the sphere rather than straight chords, the probability is 0: a simple curve on  $S^2$  is unknotted.